M3351 Series CMSIS BSP Guide

Directory Introduction for 32-bit NuMicro® Family

Please extract the “M3351_Series_BSP_CMSIS_V3.00.003.zip” file firstly and then put the “M3351_Series_BSP_CMSIS_V3.00.003” folder into the working folder.

To experience the powerful features of M3351 series in few minutes, please select the sample code to download and execute on the NuMaker board. Open the project files to build them with Keil® MDK, IAR or VS Code, and then download and trace them on the NuMaker board to see how it works.

Directory Information

Document	Driver reference guide and revision history.
Library	Driver header and source files.
SampleCode	Driver sample code.
ThirdParty	Libraries from third parties.
Tool	Utility programs.

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1 Document

SBOM	Scopes and component evidence of Product SBOM and Test Sample SBOM.
CMSIS.html	Document of CMSIS.
NuMicro M3351 Series CMSIS BSP Driver Reference Guide.chm	This document describes the usage of drivers in M3351 Series CMSIS BSP.
NuMicro M3351 Series CMSIS BSP Revision History.pdf	This document shows the revision history of M3351 Series CMSIS BSP.
VS Code Quick Start Guide.pdf	This document guides to install, configure and use VS Code.

2 Library

CMSIS	Cortex® Microcontroller Software Interface Standard (CMSIS) definitions and CMSIS-Compiler software components by Arm® Corp.
Commu	Helper functions of communication protocols, e.g., xmodem.
CryptoAccelerator	Crypto accelerator source code for MbedTLS library.
Device	CMSIS compliant device header files.
LlsiYcableLib	LLSI library source code.
StdDriver	All peripheral driver header and source files.
Storage	Disk I/O modules for FatFs.
UsbHostLib	USB host library source code.

3 SampleCode

Crypto	Crypto sample codes using MbedTLS library.
FreeRTOS	Simple FreeRTOS™ demo codes.
Hard_Fault_Sample	Show hard fault information when hard fault happened. The hard fault handler shows some information including program counter, which is the address where the processor is executed when the hard fault occurs. The listing file (or map file) can show what function and instruction that is. It also shows the Link Register (LR), which contains the return address of the last function call. It can show the status where CPU comes from to get to this point.
ISP	Sample codes for In-System-Programming.
NuMaker-M3351KI	Sample codes for NuMaker-M3351KI board.
PowerManagement	Sample codes for power management.
SecureApplication	Sample codes for secure application. VS Code projects require Python 3.12 at least for post-build.
StdDriver	Sample codes to demonstrate the usage of M3351 series MCU peripheral driver APIs.
Template	Project template for M3351 series MCU.
TrustZone	Demo of secure codes and non-secure codes.
XOM	Demonstrate how to create XOM library and use it.

4 ThirdParty

FatFs	An open-source FAT/exFAT file system library. A generic FAT file system module for small embedded systems. Its official website is https://elm-chan.org/fsw/ff/.
FreeRTOS	Real-time operating system for microcontrollers. Its official website is http://www.freertos.org/.
mbedtls	Mbed TLS offers an SSL library with an intuitive API and readable source code, so you can understand what the code does. Its official website is https://tls.mbed.org/.

5 Tool

HIDTransferTest	HID Transfer Test Tool supports two transfer modes: Interrupt Transfer Mode and Control Transfer Mode.
mcuboot	MCUboot contains the imgtool Python script and sample artifacts provided as a reference implementation for firmware signing and management. Its official website is https://github.com/mcu-tools/mcuboot/.

6 SampleCode\Crypto

mbedTLS_AES[1]	Show how mbedTLS AES function works.
mbedTLS_SHA256	Show how mbedTLS SHA256 function works.

[1] mbedTLS_AES sample code is unsupported on M3351xG because its CRYPTO does not include AES.

7 SampleCode\FreeRTOS

Blinky	This project provides two demo applications. A simple blinky style project, and a more comprehensive test and demo application.
TicklessIdle	Show how to enable FreeRTOS™ tickless idle mode, and enable RTC for long duration task idle.
TrustZone	Show how to enable FreeRTOS™ on TrustZone application.

8 SampleCode\ISP

common	Simplified system files, drivers and ISP command handle function due to LDROM code size limitation.
ISP_CAN	In-System-Programming Sample code through CAN interface.
ISP_DFU	In-System-Programming Sample code through USB interface and following Device Firmware Upgrade Class Specification.
ISP_HID	In-System-Programming Sample code through USB HID interface.
ISP_I2C	In-System-Programming Sample code through I^2C interface.
ISP_MSC	In-System-Programming Sample code through USB interface and following Mass Storage Class Specification.
ISP_RS485	In-System-Programming Sample code through RS485 interface.
ISP_SPI	In-System-Programming Sample code through SPI interface.
ISP_UART	In-System-Programming Sample code through UART interface.

9 SampleCode\NuMaker-M3351KI

xmodem	Demonstrate how to transfer data with UART Xmodem.

10 SampleCode\NuMaker-M3351SG

xmodem	Demonstrate how to transfer data with UART Xmodem.

11 SampleCode\PowerManagement

The M3351 series MCU provides some power modes with different power consumption level and wake-up time. For more information, please refer to the application note.

SYS_PowerDown_MinCurrent	Demonstrate how to minimize power consumption when entering Power-down mode.
SYS_PowerDownMode	Show how to enter different Power-down mode and wake-up by Wake-up Timer.
SYS_SPDMode_Wakeup	Show how to wake up system from SPD Power-down mode by different wakeup sources.
SYS_SPDMode_WakeupAndReturn	Show how to continue executing code after wake-up from SPD Power-down mode by SRAM data retention function.
SYS_SPDMode_WakeupVTOR	Show how to execute code after wake-up from SPD Power-down mode by VTOR function.

12 SampleCode\SecureApplication

SecureBootDemo	Demonstrate how to generate the first booting image, NuBL2. After NuBL2 runs, NuBL2 will authenticate NuBL32 and NuBL33 then jump to execute in NuBL32.
SecureISPDemo[2]	Demonstrate how to initialize a SecureISP client mode. This sample code needs to work with USBH_SecureISP sample code.
USBH_SecureISP	Demonstrate how to initialize a SecureISP server mode via USB host connected to SecureISPDemo. This sample code needs to work with SecureISPDemo sample code.

[2] SecureISPDemo and USBH_SecureISP sample codes do not support M3351xG because its CRYPTO does not include AES.

13 SampleCode\StdDriver

Analog Comparator Controller (ACMP)

ACMP_ComapreDAC[3]	Demonstrate ACMP comparison by comparing ACMP1_P0 input and DAC voltage and show the result on UART console.
ACMP_ComapreVBG	Demonstrate ACMP comparison by comparing ACMP1_P0 input and VBG voltage and show the result on UART console.
ACMP_Wakeup	Use ACMP to wake up system from Power-down mode while comparator output changes.
ACMP_WindowCompare	Show how to monitor ACMP input with window compare function.
ACMP_WindowLatch	Demonstrate how to use ACMP window latch mode.

Basic PWM Generator and Capture Timer (BPWM)

BPWM_Capture	Use BPWM0 channel 0 to capture the BPWM1 channel 2 waveform.
BPWM_DoubleBuffer	Change duty cycle and period of output waveform by BPWM double buffer function.
BPWM_OutputWaveform	Demonstrate how to use BPWM counter output waveform.
BPWM_SwitchDuty	Change duty cycle of output waveform by configured period.
BPWM_SyncStart	Demonstrate how to use BPWM counter synchronous start function.

[3] ACMP_CompareDAC sample code is unsupported on M3351xG because it lacks DAC.

Cache Controller (CACHE)

CACHE_CheckParityError	Demonstrate Cache parity error detection.
CACHE_Performance	Demonstrate the performance of CACHE.

Controller Area Network with Flexible Data-Rate (CAN FD)

CANFD_Loopback	Use CAN / CAN FD mode function to do internal loopback test.
CANFD_MonitorMode	Use CAN / CAN FD Monitor mode to monitor the CAN / CAN FD bus communication test.
CANFD_TxRx	Transmit and receive CAN / CAN FD messages through CAN interface.
CANFD_TxRxINT	An example of interrupt control using CAN / CAN FD bus communication.

Clock Controller (CLK)

CLK_ClockDetector	Show the usage of clock fail detector and clock frequency monitor function.

CRC Controller (CRC)

CRC_CCITT	Implement CRC in CRC-CCITT mode and get the CRC checksum result.
CRC_CRC8	Implement CRC in CRC-8 mode and get the CRC checksum result.
CRC_CRC32_DMA	Implement CRC in CRC-32 mode and get the CRC checksum result.
CRC_Polynomial	Demonstrate how to use polynomial mode and get the CRC checksum result.

Cryptographic Accelerator (CRYPTO)[4]

CRYPTO_AES	Show Crypto IP AES-128 ECB mode encrypt/decrypt function.
CRYPTO_AES_CCM	Demonstrate how to encrypt/decrypt data by AES CCM.
CRYPTO_AES_GCM	Demonstrate how to encrypt/decrypt data by AES GCM.
CRYPTO_HMAC	Show Crypto IP HMAC function.
CRYPTO_SHA	Use Crypto IP SHA engine to run through known answer SHA1 test vectors.

Digital to Analog Converter (DAC)[5]

DAC_ExtPinTrigger	Demonstrate how to trigger DAC conversion by external pin.
DAC_PDMA_PWMTrigger	Demonstrate how to use PDMA and trigger DAC0 by PWM.
DAC_PDMA_TimerTrigger	Demonstrate how to PDMA and trigger DAC by Timer.
DAC_PWMTrigger	Demonstrate how to trigger DAC by PWM.
DAC_SoftwareTrigger	Demonstrate how to trigger DAC conversion by software.
DAC_TimerTrigger	Demonstrate how to trigger DAC by Timer.

Data Flash Memory Controller (DFMC)[6]

DFMC_CRC32	Demonstrate how to use DFMC CRC32 ISP command to calculate the CRC32 checksum of DFMC data flash.
DFMC_EEPROM	Demonstrate how to configure, initialize, read and write

[4] CRYPTO sample codes do not support M3351xG because its CRYPTO does not include AES, HMAC and SHA1.

[5] DAC sample codes are unsupported on M3351xG because it lacks DAC.

[6] DFMC sample codes are unsupported on M3351xG because it lacks DFMC.

	DFMC EEPROM.
DFMC_NonBlocking	Show DFMC erase, read, and write functions in non-blocking mode.
DFMC_ReadAllOne	Demonstrate how to use DFMC Read-All-One ISP command to verify data flash pages are all 0xFFFFFFFF or not.
DFMC_RW	Show DFMC read flash IDs, erase, read, and write functions.

Enhanced 12-bit Analog-to-Digital Converter (EADC)

EADC_Accumulate	Demonstrate how to get accumulated conversion result.
EADC_ADINT_Trigger	Use ADINT interrupt to do the EADC continuous scan conversion.
EADC_Average	Demonstrate how to get average conversion result.
EADC_AverageCMP	Demonstrate how to compare average conversion result.
EADC_BandGap	Convert Band-gap and print conversion result.
EADC_BPWM_Trigger	Demonstrate how to trigger EADC by BPWM.
EADC_PDMA_BPWM_Trigger	Demonstrate how to trigger EADC by BPWM and transfer conversion data by PDMA.
EADC_PDMA_PWM_Trigger	Demonstrate how to trigger EADC by PWM and transfer conversion data by PDMA.
EADC_Pending_Priority	Demonstrate how to trigger multiple sample modules and got conversion results in order of priority.
EADC_PWM_Trigger	Demonstrate how to trigger EADC by PWM.
EADC_ResultMonitor	Monitor the conversion result of channel 2 by the digital compare function.
EADC_SWTRG_Trigger	Trigger EADC by writing EADC software trigger register.
EADC_TempSensor	Convert temperature sensor and print conversion result.

EADC_Timer_Trigger	Show how to trigger EADC by Timer.

External Bus Interface (EBI)[7]

EBI_NOR	Configure EBI interface to access NOR Flash connected on EBI interface.
EBI_SRAM	Configure EBI interface to access SRAM connected on EBI interface.

Enhanced Input Capture Timer (ECAP)

ECAP_GetEQEIFreq	Show how to use ECAP interface to get EQEI frequency.
ECAP_GetInputFreq	Show how to use ECAP interface to get input frequency

Enhanced LED Light Strip Interface (ELLSI)

ELLSI_Marquee	This is an ELLSI demo for marquee display in software mode. It needs to be used with WS2812 LED strip.
ELLSI_Marquee_Y_Cable	This is an ELLSI demo for marquee display in software mode. It needs to be used with AP6112Y LED strip. The AP6112Y LED strip will be initialized through manual software settings.
ELLSI_PDMA_Marquee	This is an ELLSI demo for marquee display in PDMA mode. It needs to be used with WS2812 LED strip.
ELLSI_PDMA_Marquee_Y_Cable	This is an ELLSI demo for marquee display in PDMA mode. It needs to be used with AP6112Y LED strip. The AP6112Y LED strip will be initialized through automatic hardware settings.

[7] EBI sample codes are unsupported on M3351xG because it lacks EBI.

Enhanced Quadrature Encoder Interface (EQEI)

EQEI_CompareMatch	Show the usage of EQEI compare function.
EQEI_PeriodicINT[8]	Show the usage of EQEI Unit Timer function.

Flash Memory Controller (FMC)

FMC_APWPROT	Demonstrate how to use FMC APROM Protect function.
FMC_CRC32	Demonstrate how to use FMC CRC32 ISP command to calculate the CRC32 checksum of APROM and LDROM.
FMC_DualBank	Demonstrate how dual processes work in dual bank flash architecture.
FMC_DualBankFwUpdate	Implement a firmware update mechanism based on dual bank flash architecture.
FMC_ECC	Demonstrated FMC ECC status and ECC error fault address.
FMC_ExeInSRAM	Implement a code and execute it in SRAM to program embedded Flash.
FMC_FwUpdateApplication	Firmware update sample code.
FMC_IAP	Show how to call LDROM function from APROM.
FMC_MultiBoot	Implement a multi-boot system to boot from different applications in APROM. A LDROM code and 4 APROM code are implemented in this sample code.
FMC_ReadAllOne	Demonstrate how to use FMC Read-All-One ISP command to check whether the specified APROM or LDROM region are all 0xFFFFFFFF or not.
FMC_RW	Show FMC read Flash IDs, erase, read, and write functions.

[8] EQEI_PeriodicINT sample code is unsupported on M3351xG because it lacks EQEI1.

FMC_XOM	An example of using FMC APIs to configure/erase XOM regions for dual-bank operation. To ensure XOM functions work after bank remap, XOM regions must be configured at the corresponding offset in each bank.

General Purpose I/O (GPIO)

GPIO_EINTAndDebounce	Show the usage of GPIO external interrupt function and de-bounce function.
GPIO_EINTTriggerPDMA	Show the usage of GPIO EINT trigger PDMA function.
GPIO_EINTTriggerPWM	Show the usage of GPIO EINT trigger PWM function.
GPIO_INT	Show the usage of GPIO interrupt function.
GPIO_OutputInput	Show how to set GPIO pin mode and use pin data input and output control.
GPIO_PowerDown	Show how to wake up system from Power-down mode by GPIO interrupt.

I²C Serial Interface Controller (I²C)

I2C_DoubleBuffer_Slave	Demonstrate how to set I²C two-level buffer in Slave mode to receive data from a master. This sample code needs to work with I2C_MultiBytes_Master sample code.
I2C_EEPROM	Show how to use I²C interface to access EEPROM.
I2C_Loopback	Demonstrate how to set I²C Master mode and Slave Mode and show how a master accesses a slave on a chip.
I2C_Master	Show how a master accesses a slave. This sample code needs to work with I2C_Slave sample code.
I2C_MultiBytes_Master	Demonstrate how to use multi-byte API to access slave. This sample code needs to work with I2C_Slave sample code.
I2C_PDMA_TRx	Demonstrate I²C PDMA mode and need to connect I2C0 (master) and I2C1 (slave).

I2C_Slave	Demonstrate how to set I²C in Slave mode to receive data from a master. This sample code needs to work with I2C_Master sample code.
I2C_SMBus	Demonstrate how to control SMBus interface and use SMBus protocol between Host and Slave.
I2C_Wakeup_Slave	Show how to wake up MCU from Power-down mode via the I²C interface. This sample code could work with I2C_Master sample code.

I3C Serial Interface Controller (I3C)[9]

I3C_Controller	Demonstrates how to initialize the I3C Controller and perform operations to the I3C Target. This sample code needs to use two boards.
I3C_Target	Demonstrates how to initialize the I3C Target and response command from the I3C Controller. This sample code needs to use two boards.

LED Light Strip Interface (LLSI)

LLSI_Marquee	This is a LLSI demo for marquee display in software mode. It needs to be used with WS2812 LED strip.
LLSI_PDMA_Marquee	This is a LLSI demo for marquee display in PDMA mode. It needs to be used with WS2812 LED strip.
LLSI_Y_Cable_Control	This is a LLSI demo for Y-cable control. It needs to be used with AP6112Y LED strip.

PDMA Controller (PDMA)

PDMA_BasicMode	Use PDMA0 channel 2 to transfer data from memory to memory.

[9] I3C sample codes are unsupported on M3351xG because it lacks I3C.

PDMA_ScatterGather	Use PDMA0 channel 4 to transfer data from memory to memory by scatter-gather mode.
PDMA_ScatterGather_ PingPongBuffer	Use PDMA to implement Ping-Pong buffer by scatter-gather mode (memory to memory).

PWM Generator and Capture Timer (PWM)

PWM_AccumulatorINT_TriggerPDMA	Demonstrate how to use PWM accumulator interrupt trigger PDMA.
PWM_AccumulatorStopMode	Demonstrate PWM accumulator stop mode.
PWM_Brake	Demonstrate how to use PWM brake function.
PWM_Capture	Capture the PWM1 Channel 0 waveform by PWM1 Channel 2.
PWM_DeadTime	Demonstrate how to use PWM Dead Zone function.
PWM_DoubleBuffer	Change duty cycle and period of output waveform by PWM Double Buffer function.
PWM_OutputWaveform	Demonstrate how to use PWM output waveform.
PWM_PDMA_Capture	Capture the PWM1 Channel 0 waveform by PWM1 Channel 2 and use PDMA to transfer captured data.
PWM_SwitchDuty	Change duty cycle of output waveform by configured period.
PWM_SyncStart	Demonstrate how to use PWM counter synchronous start function.

Quad Serial Peripheral Interface (QSPI)

QSPI_DualMode_Flash	Access SPI Flash using QSPI dual mode.
QSPI_PDMA_Master	Configure QSPI0 as Master mode and demonstrate how to communicate with an off-chip QSPI Slave device with FIFO mode. This sample code needs to work with QSPI_PDMA_Slave sample code.

QSPI_PDMA_Slave	Configure QSPI as Slave mode and demonstrate how to communicate with an off-chip QSPI Master device with FIFO mode. This sample code needs to work with QSPI_PDMA_Master sample code.
QSPI_QuadMode_Flash	Access SPI Flash using QSPI quad mode.
QSPI_Slave3Wire	Configure QSPI as Slave 3-wire mode and demonstrate how to communicate with an off-chip SPI Master device with FIFO mode. This sample code needs to work with SPI_MasterFIFOMode sample code.

Random Number Generator (RNG)[10]

RNG_EntropyPoll	Generate entropy from hardware entropy source.
RNG_Random	Generate random numbers.

Real Timer Clock (RTC)

RTC_Alarm_Test	Demonstrate the RTC alarm function. It sets an alarm 10 seconds after execution.
RTC_Alarm_Wakeup	Use RTC alarm interrupt event to wake up system.
RTC_Time_Display	Demonstrate the RTC function and display current time to the UART console.

Serial Peripheral Interface (SPI)

SPI_Flash	Access SPI Flash through SPI interface.
SPI_HalfDuplex	Demonstrate SPI half-duplex mode. Configure SPI0 as master mode and SPI1 as slave mode. Both SPI0 and SPI1 are half-duplex mode.

[10] RNG sample codes are unsupported on M3351xG because it lacks PRNG and TRNG.

SPI_Loopback	Implement SPI Master loop back transfer. This sample code needs to connect MISO pin and MOSI pin together. It will compare the received data with transmitted data.
SPI_MasterFIFOMode	Configure SPI as master mode and demonstrate how to communicate with an off-chip SPI slave device with FIFO mode. This sample code could work with SPI_SlaveFIFOMode sample code.
SPI_PDMA_LoopTest	SPI read/write demo in PDMA mode. Connect SPI MISO and MOSI pins. Both TX PDMA function and RX PDMA function will be enabled.
SPI_SlaveFIFOMode	Configure SPI as slave mode and demonstrate how to communicate with an off-chip SPI master device with FIFO mode. This sample code needs to work with SPI_MasterFIFOMode sample code.
SPII2S_Master	Configure SPI as I²S master mode and demonstrate how I²S works in master mode. This sample code needs to work with SPII2S_Slave sample code.
SPII2S_PDMA_Codec	An I²S demo with PDMA function connected with audio codec.
SPII2S_PDMA_Play	An I²S demo for playing data and demonstrating how I²S works with PDMA.
SPII2S_PDMA_PlayRecord	An I²S demo for playing and recording data with PDMA function.
SPII2S_PDMA_Record	An I²S demo for recording data and demonstrating how I²S works with PDMA.
SPII2S_Slave	Configure SPI as I²S slave mode and demonstrate how I²S works in slave mode. This sample code needs to work with SPII2S_Master sample code.

System Manager (SYS)

SYS_BODWakeup	Show how to wake up system from Power-down mode by brown-out detector interrupt.
SYS_PLLClockOutput	Change system clock to different PLL frequency and output

	system clock from CLKO pin.
SYS_TrimHIRC	Demonstrate how to use LXT to trim HIRC.

Timer Controller (TIMER)

TIMER_ACMPTrigger	Use ACMP to trigger Timer counter reset mode.
TIMER_CaptureCounter	Show how to use the Timer capture function to capture Timer counter value.
TIMER_Delay	Demonstrate the usage of TIMER_Delay API to generate a 1 second delay.
TIMER_EventCounter	Use TM0 pin to demonstrate Timer event counter function.
TIMER_FreeCountingMode	Use TM0_EXT pin to demonstrate Timer free counting mode function. And display the measured input frequency to UART console.
TIMER_InterTimerTriggerMode	Use TM0 pin to demonstrate inter-timer trigger mode function and display the measured input frequency to UART console.
TIMER_Periodic	Use the Timer periodic mode to generate Timer interrupt every 1 second.
TIMER_PeriodicINT	Implement Timer counting in periodic mode.
TIMER_PWM_AccumulatorINTStopMode	Demonstrate TIMER PWM accumulator interrupt to stop counting.
TIMER_PWM_AccumulatorINTTriggerPDMA	Demonstrate TIMER PWM accumulator interrupt to trigger PDMA transfer.
TIMER_PWM_ChangeDuty	Change duty cycle and period of output waveform in PWM down count type.
TIMER_PWM_OutputWaveform	Demonstrate output different duty waveform in Timer0~Timer3 PWM.
TIMER_TimeoutWakeup	Use Timer to wake up system from Power-down mode periodically.

TIMER_ToggleOut	Demonstrate the Timer toggle out function.

UART Interface Controller (UART)

UART_AutoBaudRate	Show how to use auto baud rate detection function.
UART_AutoFlow	Transmit and receive data using auto flow control.
UART_IrDA	Transmit and receive UART data in UART IrDA mode.
UART_LIN	Transmit LIN frame including header and response in UART LIN mode.
UART_PDMA	Demonstrate UART transmit and receive function with PDMA.
UART_RS485	Transmit and receive data in UART RS485 mode.
UART_SingleWire	Transmit and receive data in UART single-wire mode.
UART_TxRxFunction	Transmit and receive data from PC terminal through RS232 interface.
UART_Wakeup	Show how to wake up system from Power-down mode by UART interrupt.

USB 1.1 Device Controller (USBD)

USBD_Audio_Codec	Demonstrate how to implement a USB audio class device.
USBD_Audio_Headset	Demonstrate how to implement a USB audio class device. Codec is used in this sample code to play the audio data from Host. It also supports to record data from codec to Host.
USBD_HID_Keyboard	Demonstrate how to implement a USB keyboard device. It supports to use GPIO to simulate key input.
USBD_HID_Mouse	Show how to implement a USB mouse device. The mouse cursor will move automatically when this mouse device connecting to PC by USB.

USBD_HID_MouseKeyboard	Simulate an USB HID mouse and HID keyboard. Mouse draws circle on the screen and Keyboard uses GPIO to simulate key input.
USBD_HID_RemoteWakeup	Demonstrate how to implement a USB mouse device. It uses PA0 ~ PA5 to control mouse direction and mouse key. It also supports USB suspend and remote wakeup.
USBD_HID_Touch	Demonstrate how to implement a USB touch digitizer device. Two lines demo in Paint.
USBD_HID_Transfer	Demonstrate how to transfer data between a USB device and PC through a USB HID interface. A windows tool is also included in this sample code to connect with a USB device.
USBD_HID_Transfer_And_ Keyboard	Demonstrate how to implement a composite device of HID transfer and keyboard. Transfer data between USB device and PC through USB HID interface. A windows tool is also included in this sample code to connect with a USB device.
USBD_HID_Transfer_And_ MSC	Demonstrate how to implement a composite device of HID transfer and mass storage. Transfer data between USB device and PC through USB HID interface. A windows tool is also included in this sample code to connect with a USB device.
USBD_HID_Transfer_CTRL	Use USB host core driver and HID driver. It shows how to submit HID class request and how to read data from control pipe. A windows tool is also included in this sample code to connect with a USB device.
USBD_MassStorage_CDROM	Demonstrate the emulation of USB Mass-Storage device, CD-ROM.
USBD_MassStorage_Flash	Use Flash as storage to implement a USB Mass-Storage device.
USBD_MassStorage_SRAM	Use internal SRAM as backend storage media to simulate a USB pen drive.
USBD_MicroPrinter	Demonstrate how to implement a USB micro printer device.
USBD_Printer_And_HID_	Demonstrate how to implement a composite device of USB

Transfer	micro printer and HID transfer. Transfer data between USB device and PC through USB HID interface. A windows tool is also included in this sample code to connect with a USB device.
USBD_VCOM_And_HID_ Keyboard	Demonstrate how to implement a composite device of VCOM and HID keyboard.
USBD_VCOM_And_HID_ Transfer	Demonstrate how to implement a composite device of VCOM and HID transfer. Transfer data between USB device and PC through USB HID interface. A windows tool is also included in this sample code to connect with a USB device.
USBD_VCOM_And_Mass Storage	Demonstrate how to implement a composite device of VCOM and mass storage.
USBD_VCOM_MultiPort	Demonstrate how to implement a USB multiple virtual COM port device.
USBD_VCOM_SerialEmulator	Demonstrate how to implement a USB virtual COM port device.
USBD_VCOM_SerialEmulator _DoubleBuffer	Demonstrate how to implement a USB virtual COM port device using double buffer mode.
USBD_VENDOR_LBK	This sample code works as a proprietary Vendor LBK device. It's created for USBH_VENDOR_LBK of this BSP. Vendor LBK device includes Control, Bulk, Interrupt, and Isochronous in/out endpoint pairs. Each endpoint pair receive data from host via the out-endpoint and send data back to host via the in-endpoint.

USB Host Controller (USBH)

USBH_DEV_CONN	Use connect/disconnect callback functions to handle of device connect and disconnect events.
USBH_FirmwareUpdate	Automatically search and read new firmware from USB drive, if found, update APROM Flash with it.
USBH_HID	Use USB Host core driver and HID driver. Demonstrate how to submit HID class request and read data from

	interrupt pipe. This sample code supports dynamic device plug/un-plug and multiple HID devices.
USBH_HID_Keyboard	Demonstrate reading key inputs from USB keyboards. This sample code includes an USB keyboard driver which is based on the HID driver.
USBH_HID_MouseKeyboard	Demonstrate how to support USB mouse and keyboard input.
USBH_MassStorage	Use a command-shell-like interface to demonstrate how to use USBH mass storage driver and make it work as a disk driver under the FATFS file system.
USBH_UAC	Demonstrate how to use USBH Audio Class driver. It shows the mute, volume, auto-gain, channel, and sampling rate control.
USBH_UAC_HID	Show how to use USBH Audio Class driver and HID driver at the same time. The target device is a Game Audio (UAC+HID composite device).
USBH_UAC_Loopback	Receive audio data from an UAC device, and immediately send back to the UAC device.
USBH_VCOM	Demonstrate how to use the USB Host core driver and CDC driver to connect a CDC class VCOM device.
USBH_VCOM_MassStorage	This sample uses a command-shell-like interface to demonstrate how to use USBH mass storage driver and make it working as a disk driver under FATFS file system. It also demonstrates how to use CDC driver to connect a CDC class VCOM device.
USBH_VENDOR_LBK	Show how to do transfer on a known device with a vendor driver. This sample code requires a USB device running sample USBD_VENDOR_LBK to be connected.

Universal Serial Control Interface Controller - I²C Mode (USCI-I2C)

USCI_I2C_EEPROM	Demonstrate how to access EEPROM through a USCI_I2C interface.
USCI_I2C_Master	Demonstrate how a Master accesses Slave. This sample

	code needs to work with USCI_I2C_Slave sample code.
USCI_I2C_Master_10bit	Demonstrate how a Master uses 10-bit addressing access Slave. This sample code needs to work with USCI_I2C_Slave_10bit sample code.
USCI_I2C_Monitor	Demonstrate how USCI_I2C monitors transmission between I²C Master and I²C Slave.
USCI_I2C_MultiBytes_Master	Demonstrate how to use multi-byte API to access slave. This sample code needs to work with the USCI_I2C_Slave sample code.
USCI_I2C_Slave	Demonstrate how to set USCI_I2C in slave mode to receive the data from a Master. This sample code needs to work with USCI_I2C_Master sample code.
USCI_I2C_Slave_10bit	Demonstrate how to set USCI_I2C in 10-bit addressing slave mode to receive the data from a Master. This sample code needs to work with USCI_I2C_Master_10bit sample code.
USCI_I2C_Wakeup_Slave	Demonstrate how to set USCI_I2C to wake up MCU from Power-down mode. This sample code needs to work with USCI_I2C_Master sample code.

Universal Serial Control Interface Controller - SPI Mode (USCI-SPI)

USCI_SPI_Loopback	Implement USCI_SPI0 master loop back transfer. This sample code needs to connect USCI_SPI0_MISO pin and USCI_SPI0_MOSI pin together. It will compare the received data with transmitted data.
USCI_SPI_MasterMode	Configure USCI_SPI0 as master mode and demonstrate how to communicate with an off-chip SPI Slave device. This sample code needs to work with USCI_SPI_SlaveMode sample code.
USCI_SPI_PDMA_LoopTest	Demonstrate SPI data transfer with PDMA. USCI_SPI0 will be configured as master mode and USCI_SPI1 will be configured as slave mode. Both Tx PDMA function and Rx PDMA function will be enabled.

USCI_SPI_SlaveMode	Configure USCI_SPI0 as slave mode and demonstrate how to communicate with an off-chip SPI master device. This sample code needs to work with USCI_SPI_MasterMode sample code.

Universal Serial Control Interface Controller - UART Mode (USCI-UART)

USCI_UART_AutoBaudRate	Show how to use auto baud rate detection function.
USCI_UART_Autoflow	Transmit and receive data with auto flow control.
USCI_UART_PDMA	This is a USCI_UART PDMA demo and needs to connect USCI_UART Tx and Rx.
USCI_UART_RS485	Transmit and receive data in RS485 mode.
USCI_UART_TxRxFunction	Transmit and receive data from PC terminal through RS232 interface.
USCI_UART_Wakeup	Show how to wake up system from Power-down mode by USCI interrupt in UART mode.

Watchdog Timer (WDT)

WDT_TimeoutWakeupAndReset	Implement WDT time-out interrupt event to wake up system and generate time-out reset system event while WDT time-out reset delay period expired.

Window Watchdog Timer (WWDT)

WWDT_CompareINT	Show how to reload the WWDT counter value.
WWDT_ReloadCounter	Demonstrate how to reload the WWDT counter value without resetting the MCU.

14 SampleCode\TrustZone

HardFault	Show the hard fault usages in both secure and non-secure code.
Template	Demonstrate how to implement code for secure and non-secure.

15 SampleCode\XOM

XOMLib	Show how to build and provide the XOM library and configure/erase XOM regions for dual-bank operation. To ensure XOM functions work after bank remap, XOM regions must be configured at the corresponding offset in each bank.
XOMLibDemo	Demo how to use XOM library in dual bank.

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